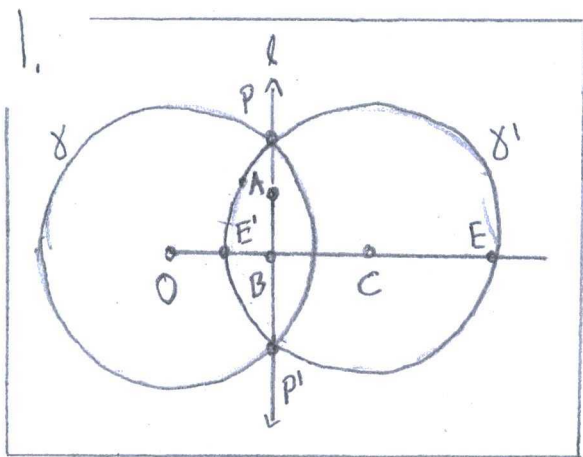


Chapter 4: Neutral Geometry:

Major Exercises



"Circle-circle continuity

(from exercise) "Circle-circle continuity implies the line-circle continuity principle."

Proof: Hypothesis

Evidence.

- 1) O is the center of X for radius OP
- 2) Line $\leftrightarrow l$ passes through a point A inside X and P .

Euclid Postulate III

RAA hypothesis

Case 1: $O = A$

- A) Line $\leftrightarrow l$ intersects X in two points: P and P' because $OP = OP'$

RAA outcome and Axiom C-1.

Case 2: $O \neq A$

- A) Point B is the foot of the perpendicular

Proposition 3.16

from O to $\tilde{l}$.

B) Point C constructs
 OC across midpoint B .

C) γ' is the circle at
 C with the same
radius as γ

D) γ' intersects OC at
point E' inside γ because $OE' < OC$

E) γ' intersects a
ray OC at a point
 E outside γ because
Segment $OC < OE$.

3) γ' intersects γ at P
and P' .

Proposition 4.3 (midpoints)

Euclid Postulate III

Proposition 3.13
(segment ordering)

Proposition 3.13.
(segment ordering)

RAA result

2. (from exercise) "Line-circle continuity principle
implies segment circle continuity
principle (or the converse)."

Proof: Hypothesis

Evidence

1) Point O is the center of γ with radius OP .

2) Line ℓ passes through a point A inside γ

Case 1: $O=A$

A) ℓ intersects γ in two points

Case 2: $O \neq A$

A) Point B is the foot of the perpendicular from O to ℓ

B) Point C constructs OC across midpoint B

C) γ' is the circle at C with same radius to γ

D) γ' intersects OC at point E' inside γ because $OE' < OC$

E) γ' intersects a

Euclid Postulate III

RAA hypothesis

RAA outcome and C-1

Proposition 3.16

Proposition 4.3
(midpoints)

Euclid Postulate III

Proposition 3.13

ray OC at point E
outside γ because
segment $OC < OE$.

Proposition 3.13
(midpoints)

3) E and E' are endpoints
inside and outside γ
with B in between.

RAA outcome.

3.

a) (from exercise) "There exists a right triangle with
a hypotenuse of length c and
a leg of length b iff $b < c$ "

Proof: Hypotenuse	Evidence
1) O is the center of γ with radius OP	Euclid's Postulate III
2) Line ℓ passes through a point A inside γ	RAA hypothesis
<u>Case 1: $O = A$</u>	
A) A right triangle is not possible, b nor c .	RAA outcome

Case 2: $O \neq A$

A) Point B is the foot of the perpendicular from O to l .

B) Length c is OP_1 and b is OB .

Proposition 3.16

RAA result.

b) (from exercise) "Whenever $b < c$, there exists a right triangle with hypotenuse of length c and a leg of length b ... implies the line-circle continuity principle."

Proof: Hypotenuse

Evidence.

1) Vertices of a triangle are P, B, O , with hypotenuse c and a leg b .

RAA guess

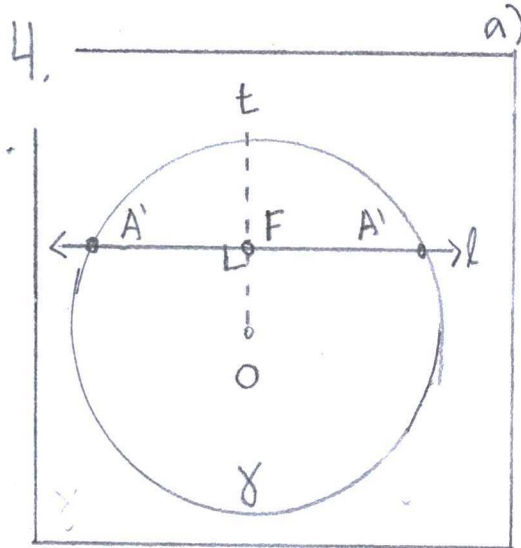
2) B is the midpoint between P and P'

Proposition 4.3 (midpoints)

3) Points P and P'

RAA facts

define the line-circle
continuity principle



"Secant"

(from exercise) "The foot F of the perpendicular t from O to l lies inside χ "

Proofs	Hypotenuse	Evidence
1) Segment OA is the radius to circle χ with center at O		Euclid's Postulate <u>III</u>
2) A line l passes through χ at A , and also A'		Line-circle continuity Principle
3) A point F is the midpoint between A and A' , so $AF = A'F$		Proposition 4.3 (midpoints)
4) A second line t		Axiom I-1

intersects both O
and F

$$5) \angle AOF \cong \angle A'OF$$

$$6) OA' = OA$$

Case 1: $O = F$

$$A) \perp \perp l$$

Case 2: $O \neq F$

$$A) \triangle AOF = \triangle A'OF$$

$$B) \angle AFO = \angle A'FO$$

$$C) \angle AFO = \angle A'FO \\ = 90^\circ$$

where F is inside
 γ

Axiom C-4

Axiom C-1

RAA ultimum

Axiom C-6 (SAS)

(pg 199) "Definition-
Corresponding
angles"

RAA ultimum

(from exercise) "The reflection of A' of A
across ℓ is another point at
which ℓ meets γ "

Proof: Hypothesis

1) With the info above

Evidence

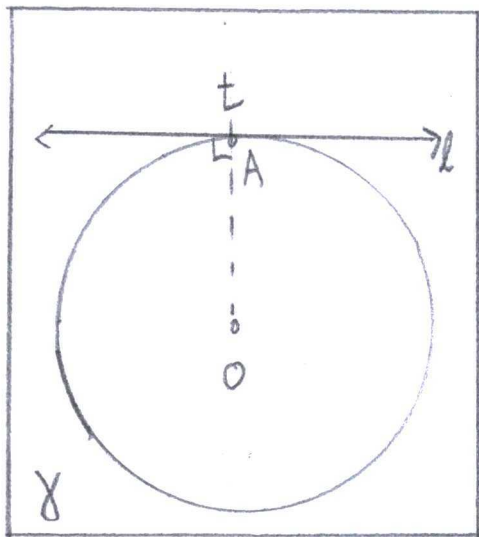
RAA premise

2) $A \neq M \neq A$

3) Point A transforms
across line ℓ onto

Axiom B-1

(pg 153-154) "Definition-
reflection"



"Tangent"

A'

b) (from exercise) "Every point $B \neq A$ lying
on ℓ is outside χ ,
hence A is a unique
point at which ℓ meets
 χ "

Proof: Hypothesis

Evidence

1) OA forms a
circle at point
O

Euclids Postulate III

2) OA is line ℓ

Axiom I-1

3) The perpendicular
line ℓ crosses
line ℓ at A

Proposition 2.5

4) Another point B
is on ℓ

Axiom I-2

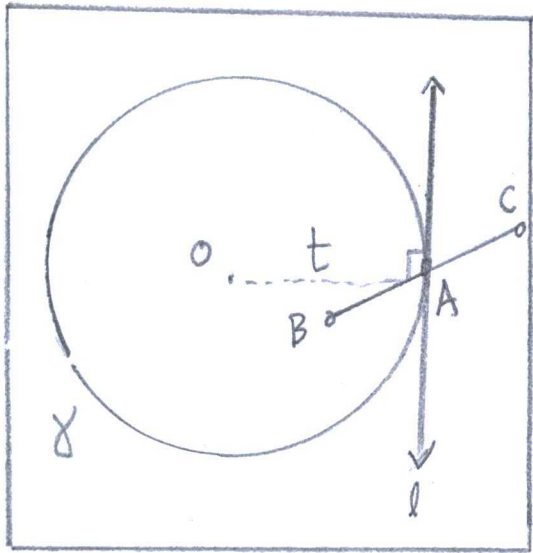
5) $OB > OA$ when $B \neq A$

Proposition 3.B

(from exercise) "All points of γ other than A are on the same side of ℓ ."

Proof: Hypothesis

Evidence



1) OA forms a circle γ at point O

Euclid's Postulate III

2) OA is a line $\vec{t}$

Axiom I-1

3) The perpendicular line ℓ crosses $\vec{t}$ at A

Proposition 2.5

4) One point B is in γ

(pg 193) "Definition - interior circle"

5) One point C is opposite B across $\vec{\ell}$

Axiom C-6.

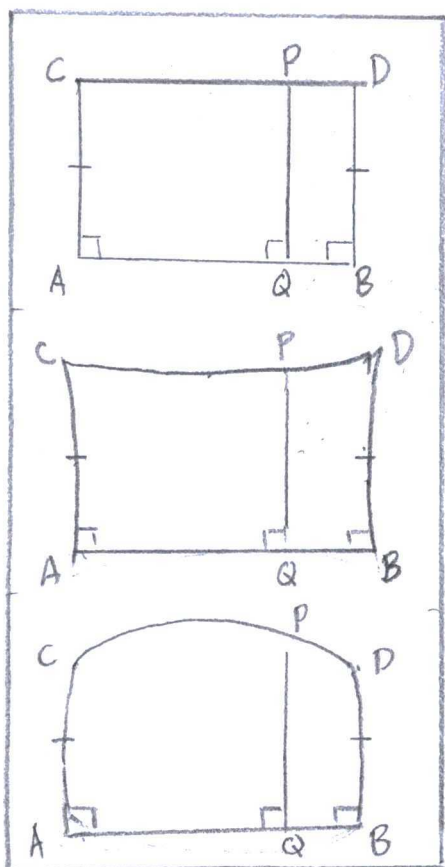
6) $B \neq A \neq C$

Axiom B-1

(from exercise) "If a line intersects a circle at only one point, then that line is tangent to the circle."

Proof	Hypothesis	Evidence
1) OA forms a circle at point O		Euclids Postulate III
2) OA is line t		Axiom I-1
3) The perpendicular line l crosses line t at A		Proposition 2.5
4) $l \perp OA$		(pg 202) "Definition-tangent."

5.



(from exercise) Lemma 1

"Given a Saccheri quadrilateral $\square ABCD$ and a point P between C and D . Let Q be the foot of the perpendicular from P to the base AB . Then

a) $PQ < BD$ iff the summit angles of $\square ABDC$ are acute

b) $PQ \equiv BD$ iff the summit angles of $\square ABCD$ are right angles

c) $PQ > BD$ iff the summit angles of

"Saccheri Quadrilaterals"

$\square ABCD$ are obtuse.

Proof: Hypothesis	Evidence
1) From circumstances provided in Lemma.	RAA guess
<u>Case 1: Acute summit angles</u>	
A) $\angle ACD = \angle BDC$	Proposition 4.12(a)
B) $\angle QPC$ is supplementary to $\angle QPD$	(pg 19) "Definition-Supplementary"
C) $PQ < BD$	Corollary to Proposition 4.13
<u>Case 2: Right summit angles</u>	
A) $\angle ACD = \angle BDC$	Proposition 4.12(a)
B) $\angle QPC$ is supplementary to $\angle QPD$	(pg 19) "Definition-Supplementary"
C) $PQ \equiv BD$	RAA result
<u>Case 3: Obtuse summit</u>	

angles

A) $\angle ACD = \angle BCD$

B) $\angle QPC$ is supplementary to

$\angle QPD$.

C) $PQ > BD$

Proposition 4.12(a)

(pg) "Definition-Supplementary"

RAA result

6.

(From exercise) Lemma 2

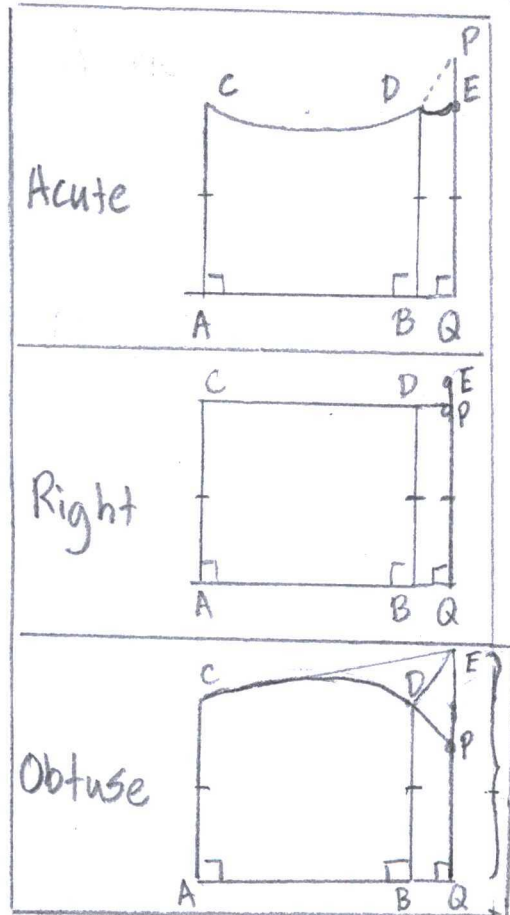
"Given a Saccheri quadrilateral $\square ABCD$ and a point P such that $C \times D \times P$.

Let Q be the foot of the perpendicular from A to AB . Then,

a) $PQ > BD$ iff the summit angles of $\square ABCD$ are acute

b) $PQ \equiv BD$ iff the summit angles of $\square ABCD$ are right angles

c) $PQ < BD$ iff the summit angles of $\square ABDC$ are obtuse.



"Saccheri
Quadrilaterals"

Proof Hypothesis

Evidence

1) From circumstances

Supplied in Lemma 2

Case 1: Acute summit
angles

A) A unique point E
is between Q
and P, so $Q * E * P$

B) $QE \cong BD$

C) $\square AQEC$ and $\square BQED$
are Saccheri quadrilaterals because
 $\angle ACD \cong \angle BDC$ and
 $\angle BDE \cong \angle QED$

D) $\angle DPQ < \angle BDP$
implies $PQ > BD$

Case 2: Right summit
angles

A) $\square AQPC$ is a
Saccheri quadrilateral
because $\angle ACP \cong \angle QPC$

RAA conjecture

(pg 70) "Definition -
collinear

Euclids Postulate II

Proposition 4.12(a)

Proposition 4.13

Proposition 4.12(a)

B) $QE \cong BD$

C) $\square AQEC$ and $\square BQED$
are Saccheri
quadrilaterals
because $\angle ACD \cong \angle BDC$
and $\angle BDE \cong \angle QED$

D) $\angle DPQ \cong \angle BDP$ implies
 $PQ \cong BD$

Case 3: Obtuse Angles

A) A unique point E
is on QP, so $Q * P * E$

B) $QE = BD$

C) Both $\square AQEC$ and
 $\square BQED$ are Saccheri
quadrilaterals because
 $\angle ACD \cong \angle QEC$ and
 $\angle BDE \cong \angle QED$

d) Exterior Angle $\angle EDP$
is greater than the
interior angle $\angle ECP$

Euclid's Postulate II

Proposition 4.12(a)

Proposition 4.13

(pg 70) "Definition-
collinear"

Measurement

Theorem 4.3(B.8)

Proposition 4.12(a)

Exterior Angle (EA)
Theorem 4.2

$$E(\angle BDE - \angle EDP = \angle BDP)$$

$$\angle(\angle ACE - \angle ECD = \angle ACD) \\ = \angle BDC]$$

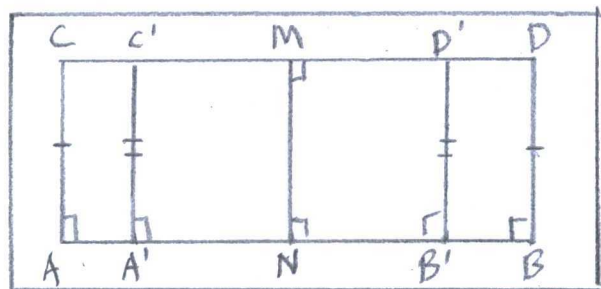
Proposition 3.20
(Angle subtraction)

F) $PQ < BD$ because

$$\angle BDP < \angle BDC$$

Proposition 4.13

7.



"Special case of
Uniformity theorem"

(pg 132) Uniformity Theorem.

"For any Hilbert plane, if one
Saccheri quadrilateral has

acute (respectively, right, obtuse)

summit angles, then so do

all Saccheri quadrilaterals."

(from exercise) Special case

"the midline segments of the two given
Saccheri quadrilaterals are congruent."

Proof: Hypothesis

Evidence

1) In a Saccheri
quadrilateral $\square ANMC$,
the segment $A'C'$ is

RAA hypothesis

perpendicular to AN

with A' the foot.

Case 1: Reflection

A) $\square ANMC$ reflects
across common side
NM for $\square MNBP$

B) $\angle AC'M = \angle B'D'M$,
So midline segments
are congruent

Case: Duplicate

A) If $\square ABDC$ duplicates,
then $M=M'$, and $N=N'$

The midpoints M, N 's
summit angles and
base lines coincide.

B) $\angle NMD' = \angle B'D'M$, so
 $MN = B'D'$

C) $\angle N'M'D' = \angle NMD$,
so $MN = M'N'$

(pg 153-154) "Definition-
reflection"

(pg 202) "Lemma 1"

Requirement 0.

(pg 202) "Lemma 2"

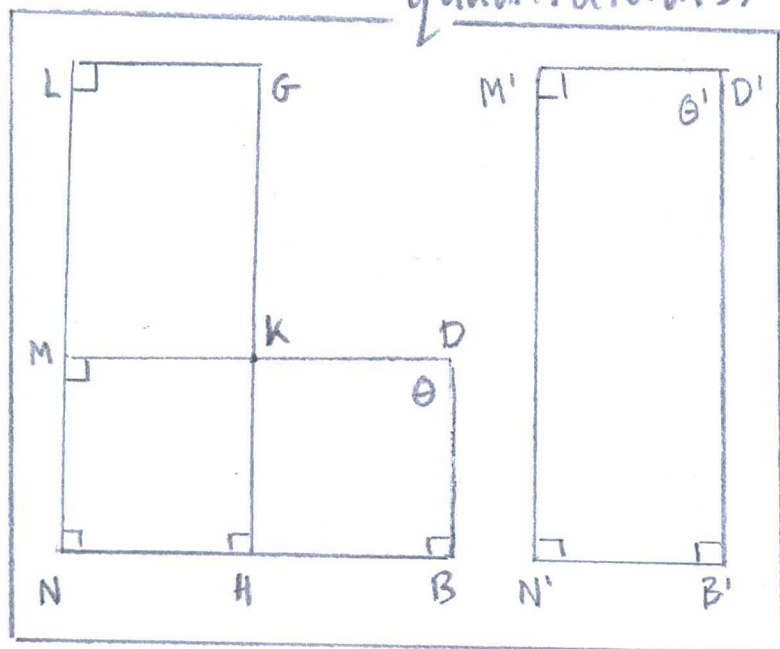
RAA result.

8.

(pg 182) Uniformity Theorem

"For any Hilbert plane, if one Saccheri

quadrilateral has acute (respectively, right, obtuse) summit angles, then so do all Saccheri quadrilaterals.



Note: Many steps in Major exercises were repetitive.

"Uniformity Theorem"

Proof by Hypothesis

Evidence

1) Two quadrilaterals resemble; $\square MNBD$ and

$\square M'N'B'D'$

2) $M'N' > MN$

3) A unique point L defines $L * M * N$ and $LN \cong MN'$

4) A Lambert quadrilateral $\square LNHG$ has a fourth

RAA hypothesis

Proposition (segment ordering)

(pg 70) "Definition - collinear"

(pg 80) "Definition - Lambert"

angle at G , congruent
to half of Saccheri
quadrilateral $\square ANB'C'$

5) On ray NB , point H
lets $NH \cong N'B'$

Euclids Postulate II

6) On the same side
of LN as H and
on the perpendicular
to LN through L ,
point G lets $LG \cong M'D'$

Euclids Postulate II

7) Then $\triangle LNH \cong \triangle M'N'B'$,
 $\angle GLH \cong \angle D'M'B'$,
 $\triangle GLH \cong \triangle D'M'B'$, so
that $\angle G \cong \angle D'$
and $\angle NHG \cong \angle B' = 90^\circ$

Axiom C-6 (S.A.S.)

8) Points G and L are
on the same side of
 MD because $GL \parallel MD$

Axiom B-4

9) Also, G and L are
on opposite sides to
 N and H across line MD

Axiom B-4

10) Point k is where GH
meets MD on the
same side of LN

Axiom B-4

$$11) \angle MKH \cong \angle D \not\cong \angle G$$

RAA result

9.

(from exercise) Passing the quotient

$$AB \cong CD \iff |AB| = |CD|$$

(from exercise) Equivalence class

$$|AB| < |CD| \iff AB < CD$$

Proof: Hypothesis

Evidence

1) Two segments appear
 $\overline{AB}$ and $\overline{CD}$ where
 $\overline{AB} < \overline{CD}$

(pg 16) "Definition-
segment"

2) Another point E forms
 $\overline{CE}$ so $\overline{AB} \cong \overline{CE}$

Euclid's Postulate II

$$3) \overline{AB} \cong \overline{CE} \iff |\overline{AB}| = |\overline{CE}|$$

"Passing the quotient"

$$4) \overline{CE} \subseteq \overline{CD}, \text{ so } |\overline{CE}| \leq |\overline{CD}|$$

(pg 149) "Definition-
convex"

$$5) |\overline{AB}| < |\overline{CD}| \iff \overline{AB} < \overline{CD}$$

Proposition 3.13(b)

(from exercise) "Trichotomy"

" $a < b$ or $a = b$ or $b < a$, and only one of these possibilities occurs."

Please review Chapter 3: Hilbert's Axioms
Exercise #22

(from exercise) Transitivity

" $a < b$ and $b < c \Leftrightarrow a < c$ "

Please review Chapter 3: Hilbert's Axioms
Exercise #23

(from exercise) "commutative"

" $a + b = b + a$ "

Proof by induction: ① Base case - $f(x) = f(x+0)$
 $= f(0+x)$

For all x , $f(x) = f(0) + f(x)$

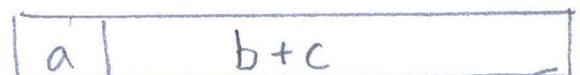
② First step - $f(x+1) = f(1+x)$
 $= f(1) + f(x)$

③ Inductive Step - $f(x+y) = f(y+x)$
 $= f(y) + f(x)$

(from exercise) "Associative"

$$(a+b)+c = a+(b+c)$$

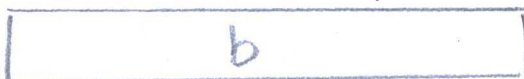
Proof by Picture



(from exercise) Subtraction

" $a < b$ iff there is a class c such that $b = a + c$ "

Proof by Picture:



(from exercise) Cancellation

" $a + c = b + c$ iff $a = b$ "

Proof by Picture:



(from exercise) Compatibility of '+' and '<'

"If $a < b$, then for any congruence class c , $a + c < b + c$ "

Proof by induction: Base case: $a < b$

First step: $a+1 < b+1$

Inductive: $a+c < b+c$
Step

(from exercise) Collinear

"If A, B, C are collinear, then $A * B * C$ "

$$A * B * C \Leftrightarrow |AB| + |BC| = |AC|$$

Proof: Hypothesis	Evidence
1) A, B, C are collinear on $A * B * C$	(pg 70) "Definition- collinear"
2) A', B', C' are collinear on $A' * B' * C'$	(pg 70) "Definition- collinear"
3) $AB \cong A'B'$	Axiom C-1
4) $BC \cong B'C'$	Axiom C-1
5) $AB + BC \cong A'B' + B'C'$	RAA conjecture
6) $ AB + BC \cong A'B' + B'C' $	"Passing the quotient"
7) $AC \cong A'C'$	Axiom C-3
8) $ AC \leq A'C' $	(pg) "Definition- subset"
9) $ AB + BC = AC $	RAA outcome